

Tugas Algoritma Genetika — Studi Kasus TSP

Mata Kuliah: Kecerdasan Komputasional

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Kelompok 6

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1 Pendahuluan

Studi kasus ini menyelesaikan *Traveling Salesperson Problem* (TSP) menggunakan Algoritma Genetika, bukan optimasi fungsi x^2 . Data yang diproses berupa koordinat dari 15 kota di Amerika Serikat. Algoritma genetika diimplementasikan dengan spesifikasi:

- **Seleksi:** Rank Selection
- **Crossover:** Order Crossover (OX)
- **Mutasi:** Inversion Mutation
- **Elitism:** Mempertahankan individu terbaik di setiap generasi
- **Jumlah Iterasi:** 5
- **Ukuran Populasi:** 4 Kromosom (untuk mempermudah perhitungan manual)
- **Peluang Crossover** (P_c): 0.8
- **Peluang Mutasi** (P_m): 0.2

2 Data Kota

Berikut adalah daftar 15 kota beserta indeksnya yang digunakan sebagai representasi gen pada kromosom.

Indeks	Kota	Latitude	Longitude
0	Denver	39.7420	-104.9915
1	Colorado Springs	38.8461	-104.8006
2	Telluride	37.9375	-107.8123
3	Las Vegas	36.1146	-115.1728
4	Grand Canyon	36.0565	-112.1251
5	Yellowstone NP	44.4237	-110.5885
6	Mount Rushmore	43.9686	-103.3818
7	Seattle	47.6080	-122.3352
8	Redwood NP	41.2131	-124.0046
9	San Diego	32.7157	-117.1610
10	Los Angeles	34.0522	-118.2437
11	Mount Hood NF	45.4543	-121.9331
12	Santa Fe	35.6915	-105.9442
13	Chicago	41.8818	-87.6232
14	New York City	40.7306	-73.9352

3 Perhitungan Manual (5 Iterasi)

Iterasi 1

1. Evaluasi Fitness

$$Fitness = 1/TotalJarak$$

- K1: [8, 13, 7, 6, 14, 12, 5, 2, 9, 3, 4, 11, 0, 1, 10] → Jarak: 243.38, Fitness: 0.004109
- K2: [2, 12, 9, 7, 4, 6, 5, 10, 13, 8, 11, 3, 1, 14, 0] → Jarak: 237.92, Fitness: 0.004203
- K3: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9] → Jarak: 206.82, Fitness: 0.004835
- K4: [12, 8, 6, 2, 10, 11, 3, 14, 7, 0, 4, 9, 13, 5, 1] → Jarak: 270.74, Fitness: 0.003694

2. Rank Selection

Kromosom diurutkan berdasarkan fitness (tertinggi ke terendah) dan diberi rank: Ter-

Rank	Kromosom	Probabilitas	Kumulatif Probabilitas
4	[7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]	0.40	0.40
3	[2, 12, 9, 7, 4, 6, 5, 10, 13, 8, 11, 3, 1, 14, 0]	0.30	0.70
2	[8, 13, 7, 6, 14, 12, 5, 2, 9, 3, 4, 11, 0, 1, 10]	0.20	0.90
1	[12, 8, 6, 2, 10, 11, 3, 14, 7, 0, 4, 9, 13, 5, 1]	0.10	1.00

pilih Parent 1: [2, 12, 9, 7, 4, 6, 5, 10, 13, 8, 11, 3, 1, 14, 0]
 Terpilih Parent 2: [8, 13, 7, 6, 14, 12, 5, 2, 9, 3, 4, 11, 0, 1, 10]

3. Crossover (Order Crossover / OX)

Menghasilkan keturunan menggunakan probabilitas crossover $P_c = 0.8$.

Elitism diterapkan: Kromosom terbaik diteruskan tanpa diubah: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]

- $r_{cross} = 0.05 \leq 0.8$: Crossover pada titik 3 hingga 12.

Anak dihasilkan: [2, 9, 0, 7, 4, 6, 5, 10, 13, 8, 11, 3, 1, 14, 12]

- $r_{cross} = 0.29 \leq 0.8$: Crossover pada titik 1 hingga 13.

Anak dihasilkan: [0, 12, 9, 7, 4, 6, 5, 10, 13, 8, 11, 3, 1, 14, 2]

- $r_{cross} = 0.23 \leq 0.8$: Crossover pada titik 1 hingga 6.

Anak dihasilkan: [14, 12, 9, 7, 4, 6, 5, 2, 3, 11, 0, 1, 10, 8, 13]

4. Mutasi (Inversion Mutation)

Mengecek kromosom baru (kecuali individu elit) untuk mutasi dengan $P_m = 0.2$.

- Anak 2 ($r_{mut} = 0.28 > 0.2$): Tidak terjadi mutasi.
- Anak 3 ($r_{mut} = 0.64 > 0.2$): Tidak terjadi mutasi.
- Anak 4 ($r_{mut} = 0.36 > 0.2$): Tidak terjadi mutasi.

Iterasi 2

1. Evaluasi Fitness

$Fitness = 1/TotalJarak$

- K1: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9] → Jarak: 206.82, Fitness: 0.004835
- K2: [2, 9, 0, 7, 4, 6, 5, 10, 13, 8, 11, 3, 1, 14, 12] → Jarak: 252.34, Fitness: 0.003963
- K3: [0, 12, 9, 7, 4, 6, 5, 10, 13, 8, 11, 3, 1, 14, 2] → Jarak: 242.08, Fitness: 0.004131
- K4: [14, 12, 9, 7, 4, 6, 5, 2, 3, 11, 0, 1, 10, 8, 13] → Jarak: 212.75, Fitness: 0.004700

2. Rank Selection

Kromosom diurutkan berdasarkan fitness (tertinggi ke terendah) dan diberi rank: Ter-

Rank	Kromosom	Probabilitas	Kumulatif Probabilitas
4	[7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]	0.40	0.40
3	[14, 12, 9, 7, 4, 6, 5, 2, 3, 11, 0, 1, 10, 8, 13]	0.30	0.70
2	[0, 12, 9, 7, 4, 6, 5, 10, 13, 8, 11, 3, 1, 14, 2]	0.20	0.90
1	[2, 9, 0, 7, 4, 6, 5, 10, 13, 8, 11, 3, 1, 14, 12]	0.10	1.00

pilih Parent 1: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]

Terpilih Parent 2: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]

3. Crossover (Order Crossover / OX)

Menghasilkan keturunan menggunakan probabilitas crossover $P_c = 0.8$.

Elitism diterapkan: Kromosom terbaik diteruskan tanpa diubah: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]

- $r_{cross} = 0.27 \leq 0.8$: Crossover pada titik 10 hingga 14.

Anak dihasilkan: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]

- $r_{cross} = 0.65 \leq 0.8$: Crossover pada titik 9 hingga 10.

Anak dihasilkan: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]

- $r_{cross} = 0.17 \leq 0.8$: Crossover pada titik 3 hingga 11.

Anak dihasilkan: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]

4. Mutasi (Inversion Mutation)

Mengecek kromosom baru (kecuali individu elit) untuk mutasi dengan $P_m = 0.2$.

- Anak 2 ($r_{mut} = 0.16 \leq 0.2$): Mutasi inversi segmen dari titik 4 hingga 6.

Hasil: [7, 8, 3, 11, 1, 14, 10, 13, 5, 6, 2, 0, 12, 4, 9]

- Anak 3 ($r_{mut} = 0.99 > 0.2$): Tidak terjadi mutasi.

- Anak 4 ($r_{mut} = 0.64 > 0.2$): Tidak terjadi mutasi.

Iterasi 3

1. Evaluasi Fitness

$Fitness = 1/TotalJarak$

- K1: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9] → Jarak: 206.82, Fitness: 0.004835
- K2: [7, 8, 3, 11, 1, 14, 10, 13, 5, 6, 2, 0, 12, 4, 9] → Jarak: 227.36, Fitness: 0.004398
- K3: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9] → Jarak: 206.82, Fitness: 0.004835
- K4: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9] → Jarak: 206.82, Fitness: 0.004835

2. Rank Selection

Kromosom diurutkan berdasarkan fitness (tertinggi ke terendah) dan diberi rank: Terpilih

Rank	Kromosom	Probabilitas	Kumulatif Probabilitas
4	[7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]	0.40	0.40
3	[7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]	0.30	0.70
2	[7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]	0.20	0.90
1	[7, 8, 3, 11, 1, 14, 10, 13, 5, 6, 2, 0, 12, 4, 9]	0.10	1.00

Parent 1: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]

Terpilih Parent 2: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]

3. Crossover (Order Crossover / OX)

Menghasilkan keturunan menggunakan probabilitas crossover $P_c = 0.8$.

Elitism diterapkan: Kromosom terbaik diteruskan tanpa diubah: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]

- $r_{cross} = 0.84 > 0.8$: Tidak terjadi crossover. Anak menyalin Parent 1.

- $r_{cross} = 0.78 \leq 0.8$: Crossover pada titik 3 hingga 13.

Anak dihasilkan: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]

- $r_{cross} = 0.03 \leq 0.8$: Crossover pada titik 5 hingga 6.

Anak dihasilkan: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]

4. Mutasi (Inversion Mutation)

Mengecek kromosom baru (kecuali individu elit) untuk mutasi dengan $P_m = 0.2$.

- Anak 2 ($r_{mut} = 0.27 > 0.2$): Tidak terjadi mutasi.

- Anak 3 ($r_{mut} = 0.21 > 0.2$): Tidak terjadi mutasi.

- Anak 4 ($r_{mut} = 0.94 > 0.2$): Tidak terjadi mutasi.

Iterasi 4

1. Evaluasi Fitness

$Fitness = 1/TotalJarak$

- K1: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9] → Jarak: 206.82, Fitness: 0.004835
- K2: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9] → Jarak: 206.82, Fitness: 0.004835
- K3: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9] → Jarak: 206.82, Fitness: 0.004835
- K4: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9] → Jarak: 206.82, Fitness: 0.004835

2. Rank Selection

Kromosom diurutkan berdasarkan fitness (tertinggi ke terendah) dan diberi rank: Terpilih

Rank	Kromosom	Probabilitas	Kumulatif Probabilitas
4	[7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]	0.40	0.40
3	[7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]	0.30	0.70
2	[7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]	0.20	0.90
1	[7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]	0.10	1.00

Parent 1: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]

Terpilih Parent 2: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]

3. Crossover (Order Crossover / OX)

Menghasilkan keturunan menggunakan probabilitas crossover $P_c = 0.8$.

Elitism diterapkan: Kromosom terbaik diteruskan tanpa diubah: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]

- $r_{cross} = 0.66 \leq 0.8$: Crossover pada titik 6 hingga 10.

Anak dihasilkan: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]

- $r_{cross} = 0.46 \leq 0.8$: Crossover pada titik 2 hingga 4.

Anak dihasilkan: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]

- $r_{cross} = 0.25 \leq 0.8$: Crossover pada titik 8 hingga 14.

Anak dihasilkan: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]

4. Mutasi (Inversion Mutation)

Mengecek kromosom baru (kecuali individu elit) untuk mutasi dengan $P_m = 0.2$.

- Anak 2 ($r_{mut} = 0.26 > 0.2$): Tidak terjadi mutasi.
- Anak 3 ($r_{mut} = 0.58 > 0.2$): Tidak terjadi mutasi.
- Anak 4 ($r_{mut} = 0.90 > 0.2$): Tidak terjadi mutasi.

Iterasi 5

1. Evaluasi Fitness

$Fitness = 1/TotalJarak$

- K1: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9] → Jarak: 206.82, Fitness: 0.004835
- K2: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9] → Jarak: 206.82, Fitness: 0.004835
- K3: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9] → Jarak: 206.82, Fitness: 0.004835
- K4: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9] → Jarak: 206.82, Fitness: 0.004835

2. Rank Selection

Kromosom diurutkan berdasarkan fitness (tertinggi ke terendah) dan diberi rank: Terpilih

Rank	Kromosom	Probabilitas	Kumulatif Probabilitas
4	[7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]	0.40	0.40
3	[7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]	0.30	0.70
2	[7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]	0.20	0.90
1	[7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]	0.10	1.00

Parent 1: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]

Terpilih Parent 2: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]

3. Crossover (Order Crossover / OX)

Menghasilkan keturunan menggunakan probabilitas crossover $P_c = 0.8$.

Elitism diterapkan: Kromosom terbaik diteruskan tanpa diubah: [7, 8, 3, 11, 10, 14,

1, 13, 5, 6, 2, 0, 12, 4, 9]

- $r_{cross} = 1.00 > 0.8$: Tidak terjadi crossover. Anak menyalin Parent 1.

- $r_{cross} = 0.51 \leq 0.8$: Crossover pada titik 1 hingga 12.

Anak dihasilkan: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]

- $r_{cross} = 0.05 \leq 0.8$: Crossover pada titik 1 hingga 2.

Anak dihasilkan: [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]

4. Mutasi (Inversion Mutation)

Mengecek kromosom baru (kecuali individu elit) untuk mutasi dengan $P_m = 0.2$.

- Anak 2 ($r_{mut} = 0.63 > 0.2$): Tidak terjadi mutasi.

- Anak 3 ($r_{mut} = 0.79 > 0.2$): Tidak terjadi mutasi.

- Anak 4 ($r_{mut} = 0.42 > 0.2$): Tidak terjadi mutasi.

4 Kesimpulan

Setelah 5 iterasi Algoritma Genetika dengan Elitism, performa program terpusat pada solusi kromosom elitnya akibat sifat konvergensi Rank Selection di populasi kecil.

Total Jarak Terpendek Terbaik: 206.82

Representasi Kromosom (Rute Terbaik): [7, 8, 3, 11, 10, 14, 1, 13, 5, 6, 2, 0, 12, 4, 9]

Rute Kota: Seattle → Redwood NP → Las Vegas → Mount Hood NF → Los Angeles → New York City → Colorado Springs → Chicago → Yellowstone NP → Mount Rushmore → Telluride → Denver → Santa Fe → Grand Canyon → San Diego

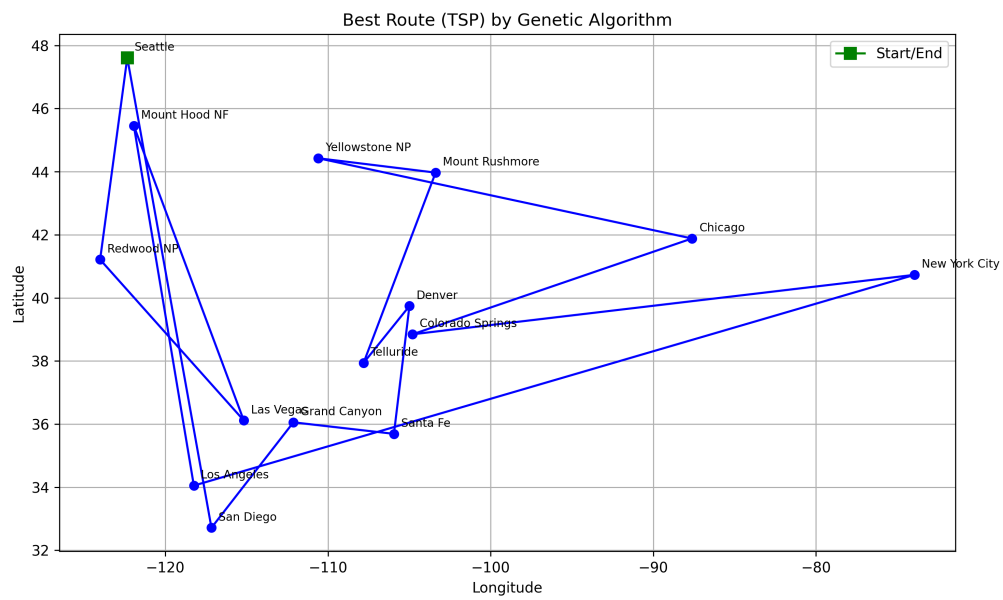


Figure 1: Visualisasi Rute Terbaik Hasil Algoritma Genetika (*Best Route*)